CSCI3150 Introduction to Operating Systems

Lecture 17: Networking

Hong Xu

https://github.com/henryhxu/CSCI3150

Agenda

- Layering
- Transport layer

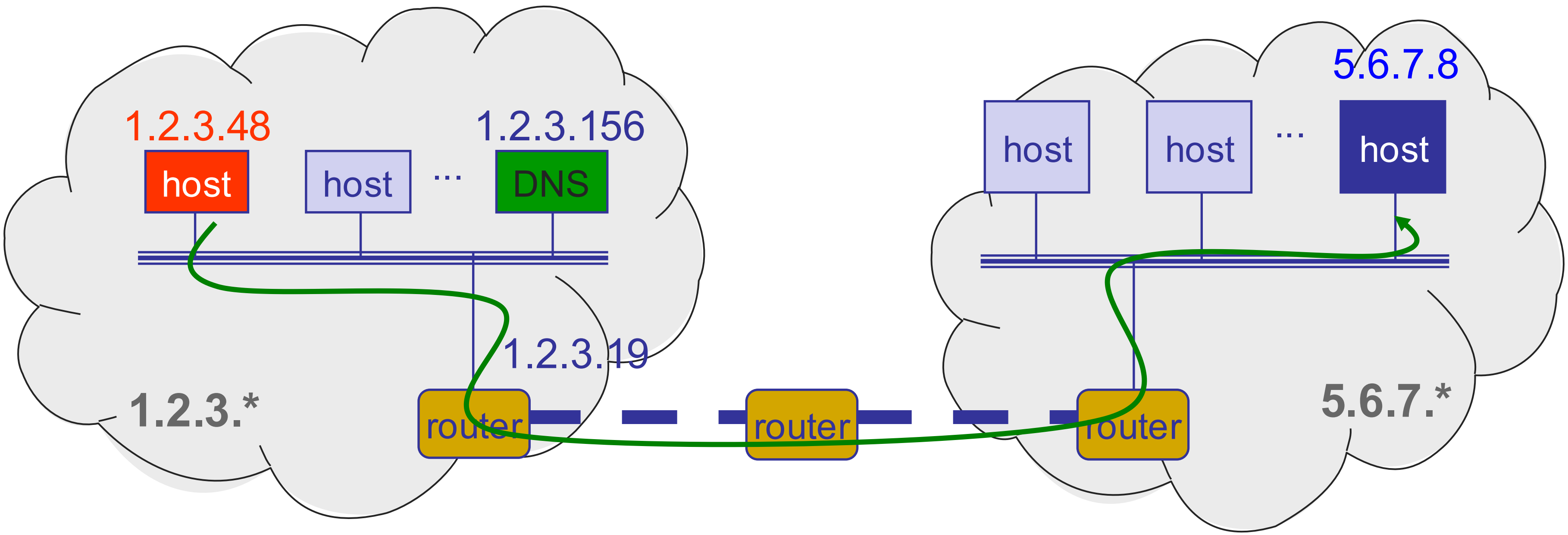
- This is just a rough introduction to the topic

Layering

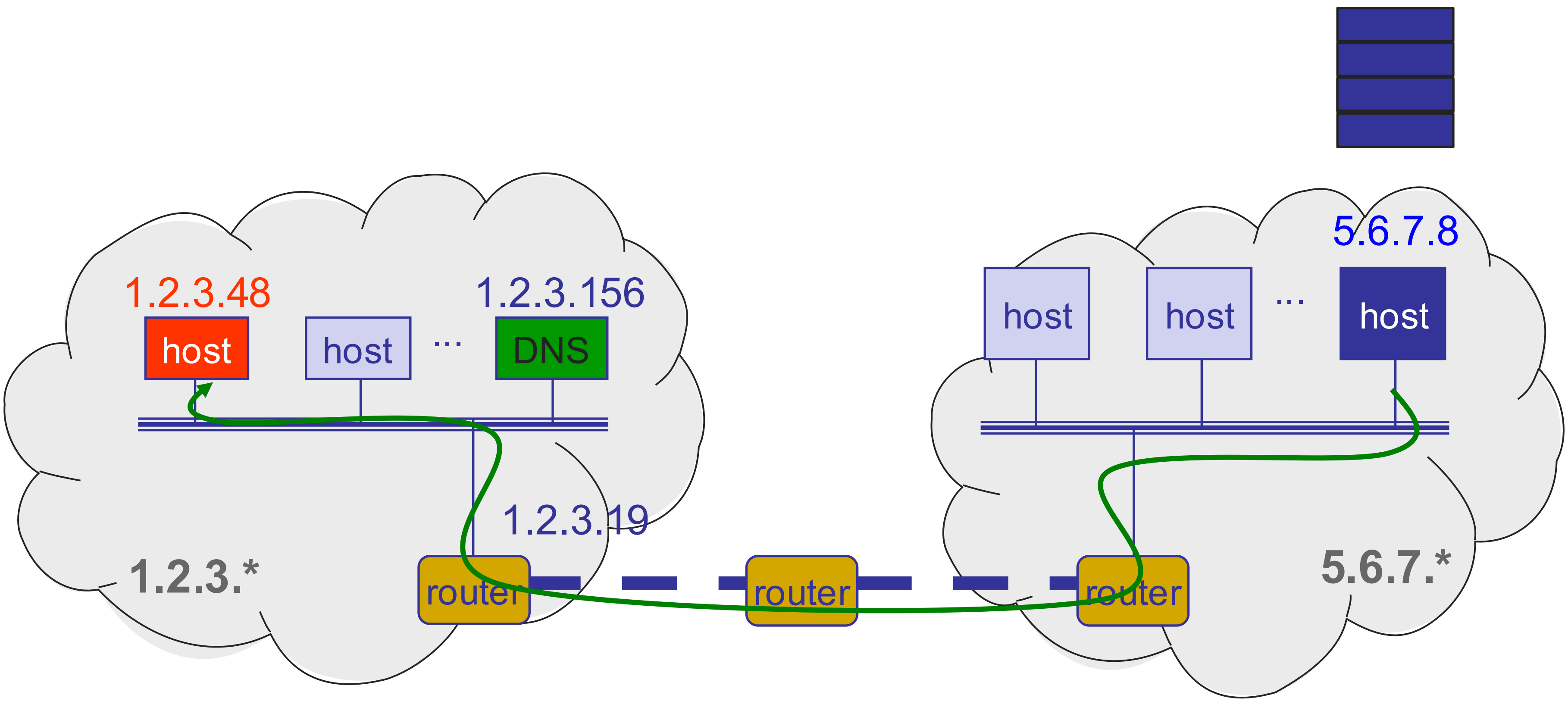
How should we organize communication?

What we want

(Some of) What happens…



(More of) What happens



What we get

Three steps

- Decompose the problem into tasks
- Organize these tasks
- Assign tasks to entities (who does what)

Back to the Internet: Decomposition

Applications

in built on

Reliable or unreliable transport

in built on

Best-effort global packet delivery

in built on

Best-effort local packet delivery

in built on

Physical transfer of bits

Communication organization

Applications

in built on

Reliable or unreliable transport

in built on

Best-effort global packet delivery

in built on

Best-effort local packet delivery

in built on

Physical transfer of bits

L7	Application
L4	Transport
L3	Network
L2	Data link
L1	Physical

OSI layers

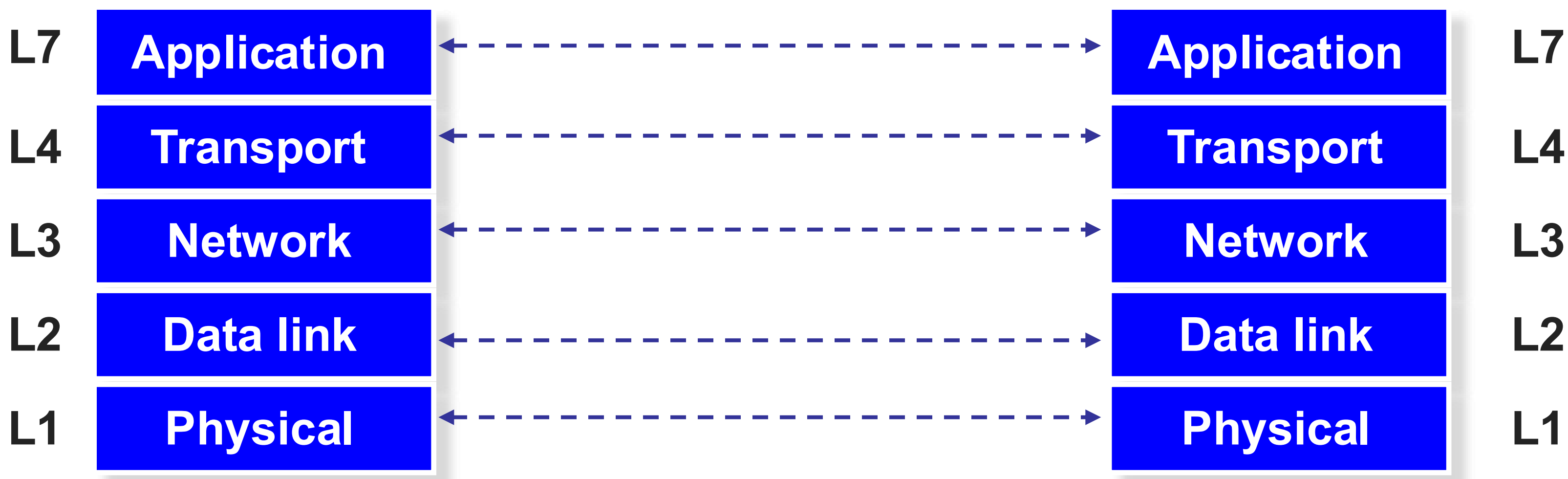
- OSI stands for Open Systems Interconnection model
  - Developed by the ISO
- Session and presentation layers are often implemented as part of the application layer

L7	Application
L6	Presentation
L5	Session
L4	Transport
L3	Network
L2	Data link
L1	Physical

Layers

- Layer: a part of a system with well-defined interfaces to other parts
- One layer interacts only with layer above and layer below
- Two layers interact only through the interface between them

Layers and protocols



- Communication between peer layers on different systems is defined by protocols

What is a Protocol?

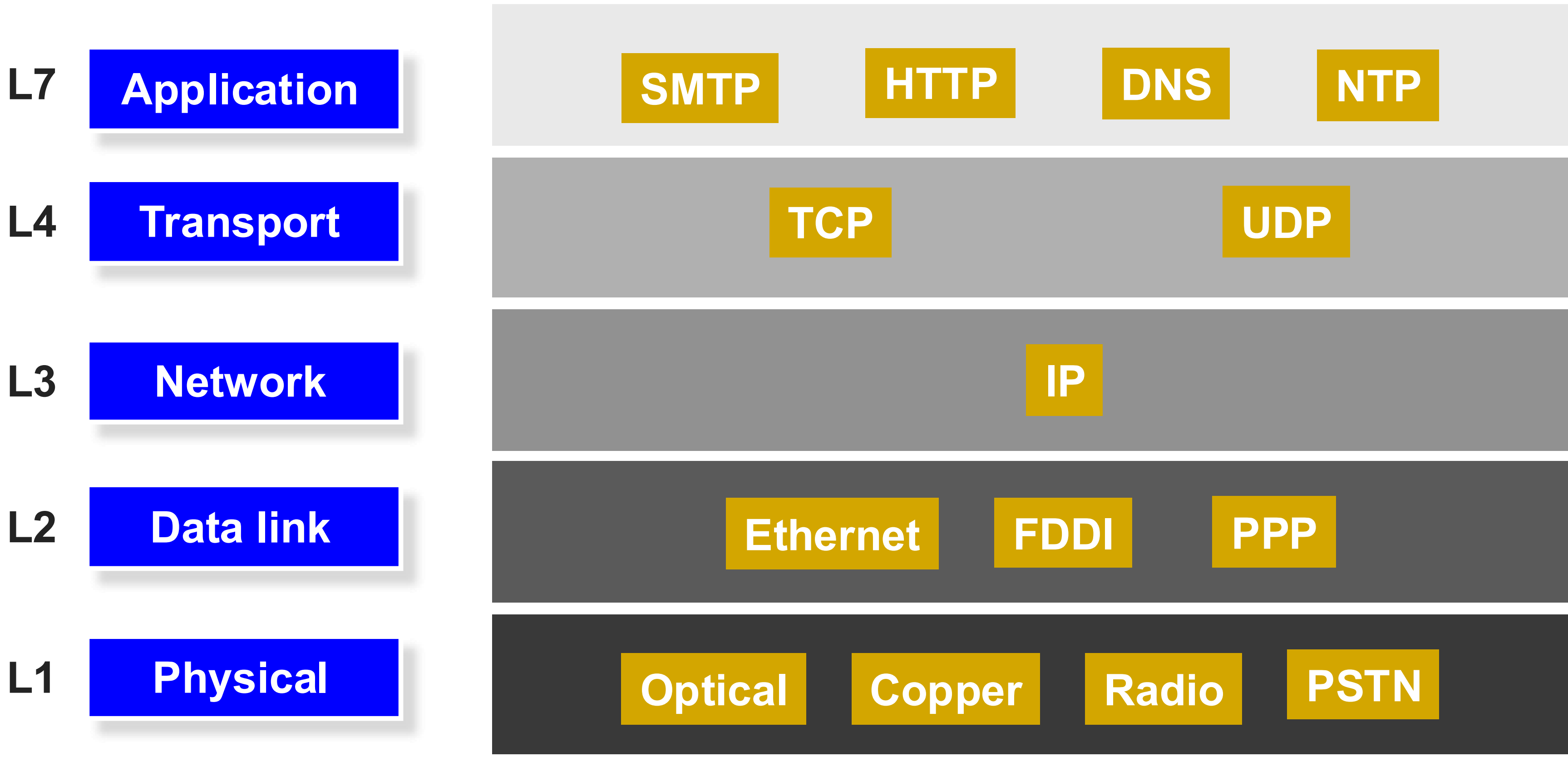
- An agreement between parties (in the same layer) on how to communicate
- Defines the syntax of communication
  - Header → instructions on how to process packet
  - Each protocol defines the format of its headers
    - e.g., “the first 32 bits carry the destination address”

Header	Payload

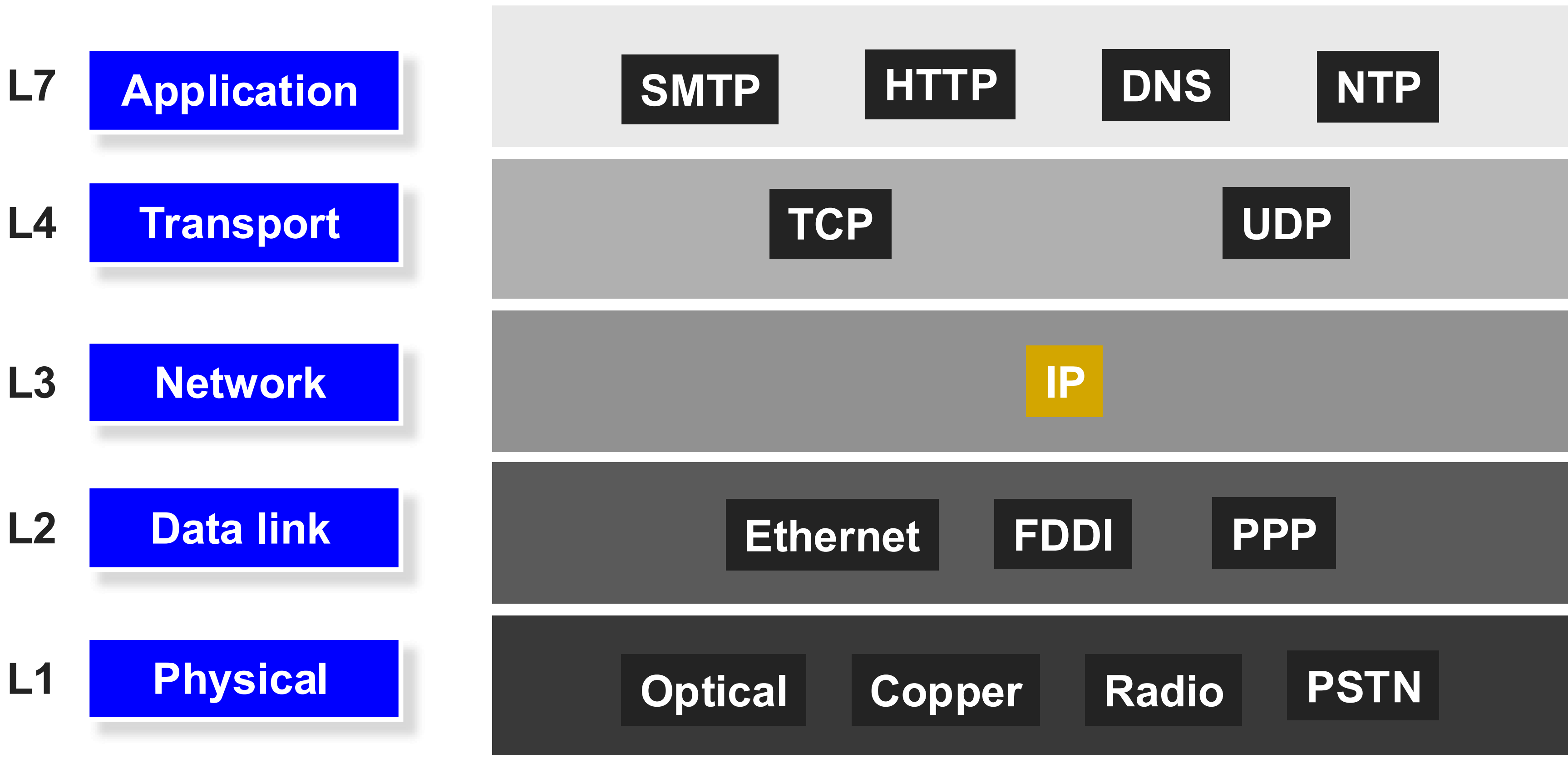
What is a Protocol?

- An agreement between parties on how to communicate
- Defines the syntax of communication
- And semantics
  - "First a hello, then a request…"
  - We will study many protocols later in the semester
- Protocols exist at many levels, hardware, and software
  - Defined by standards bodies like IETF, IEEE, ITU

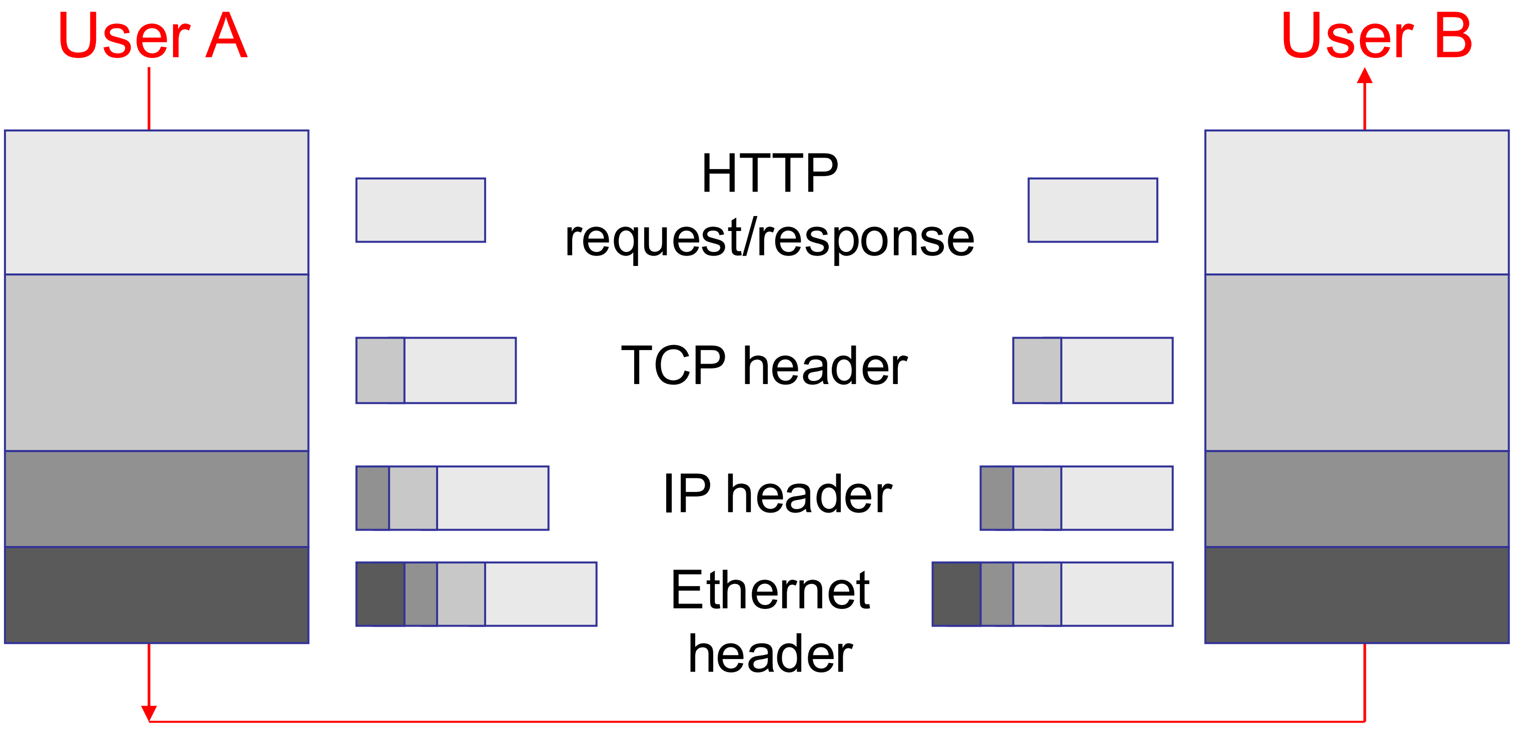
Protocols at different layers



ONE network layer protocol



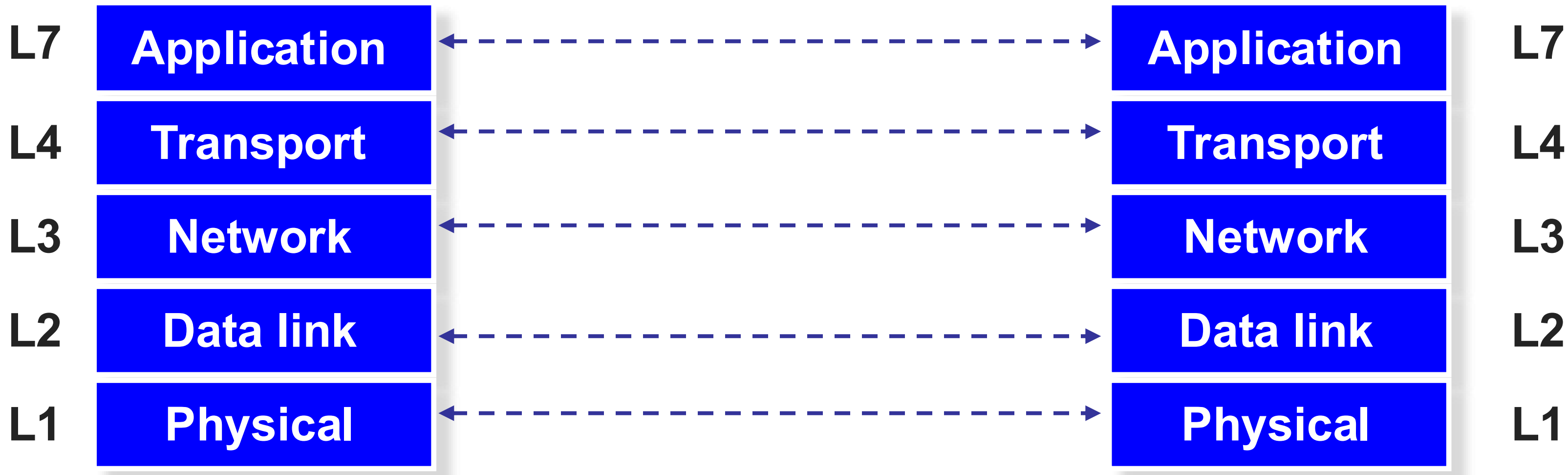
Layer encapsulation: Protocol headers



Three steps

- Decompose the problem into tasks
- Organize these tasks
- Assign tasks to entities (who does what)

What gets implemented where?



What gets implemented at the end systems?

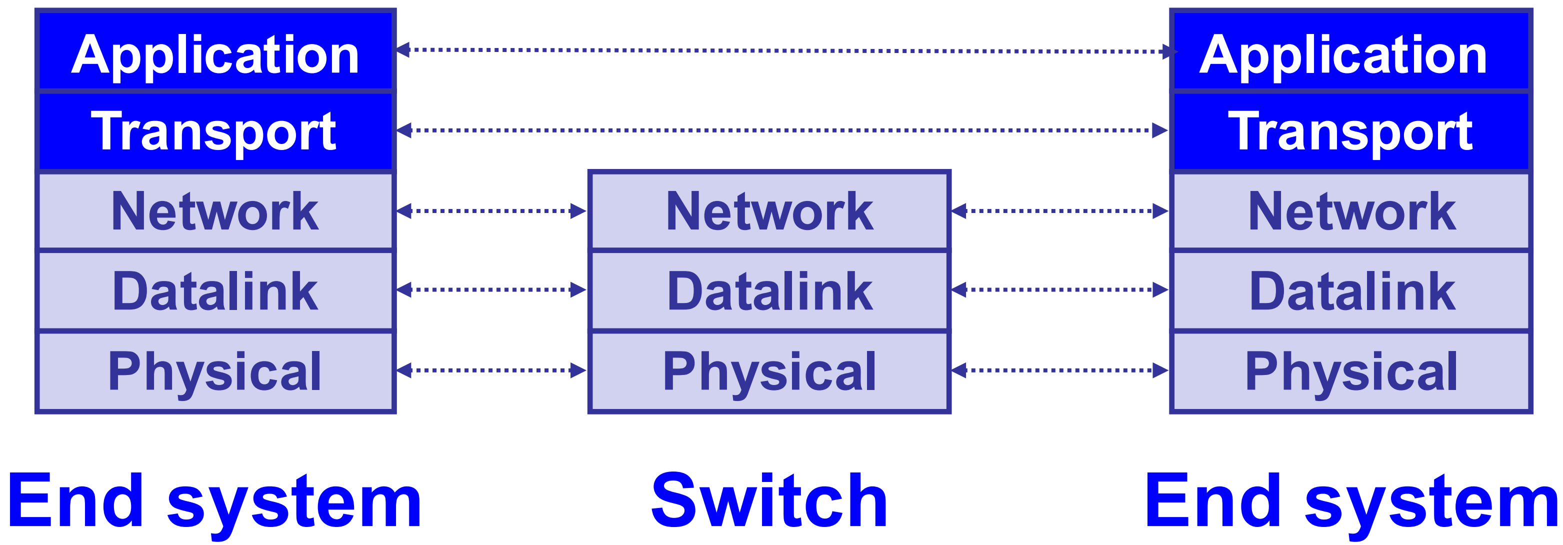
- Bits arrive on wire, must make it up to application
- Therefore, all layers must exist at host!

What gets implemented in the network?

- Bits arrive on wire → physical layer (L1)
- Packets must be delivered across links and local networks → datalink layer (L2)
- Packets must be delivered between networks for global delivery → network layer (L3)
- The network does not support reliable delivery
  - Transport layer (and above) not supported

Simple Diagram

- Lower three layers implemented everywhere
- Top two layers implemented only at hosts



A closer look: End system

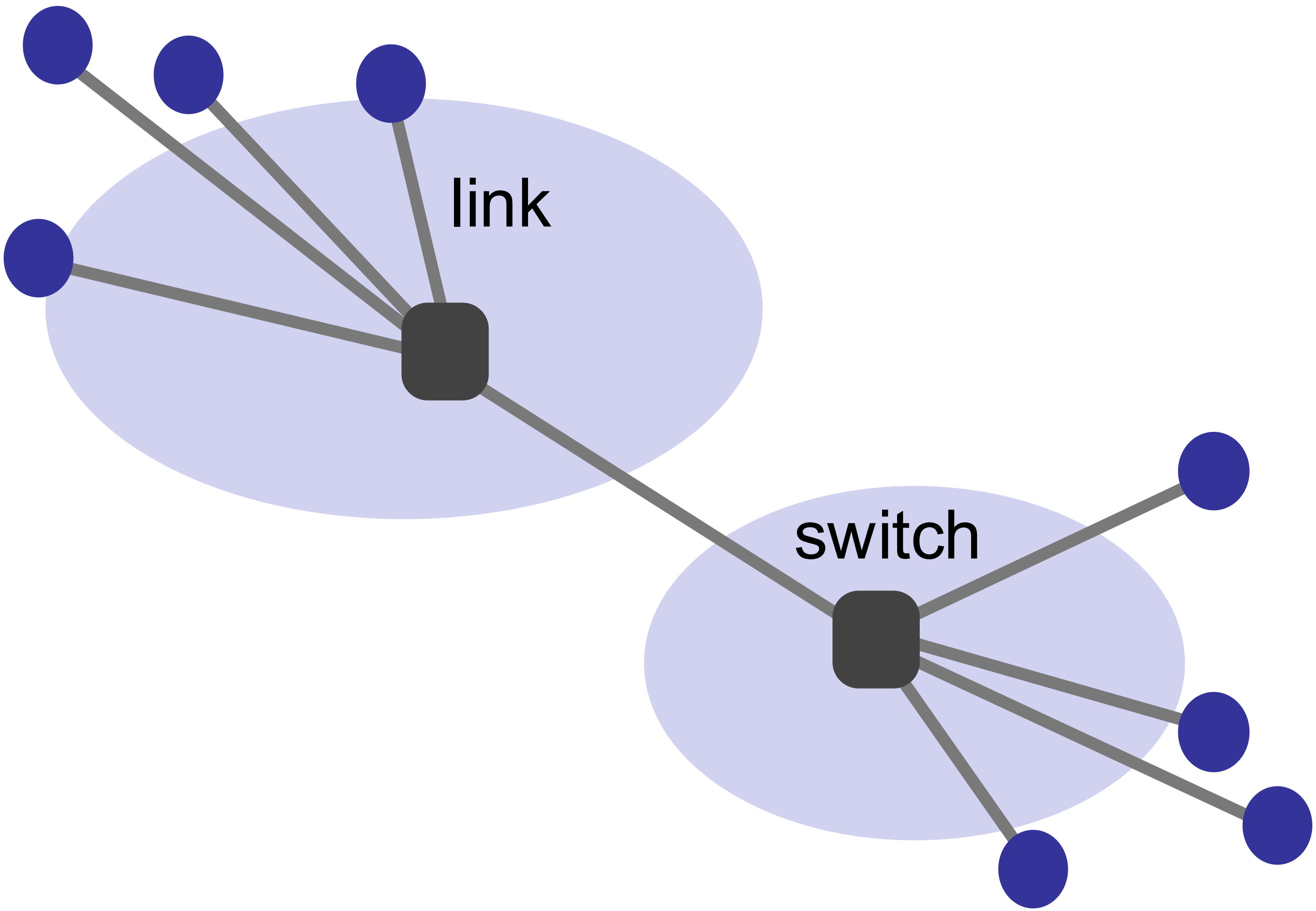
- Application
  - Web server, browser, mail, game
- Transport and network layer
  - Typically part of the operating system
- Datalink and physical layer
  - hardware/firmware/drivers

What gets implemented in the network?

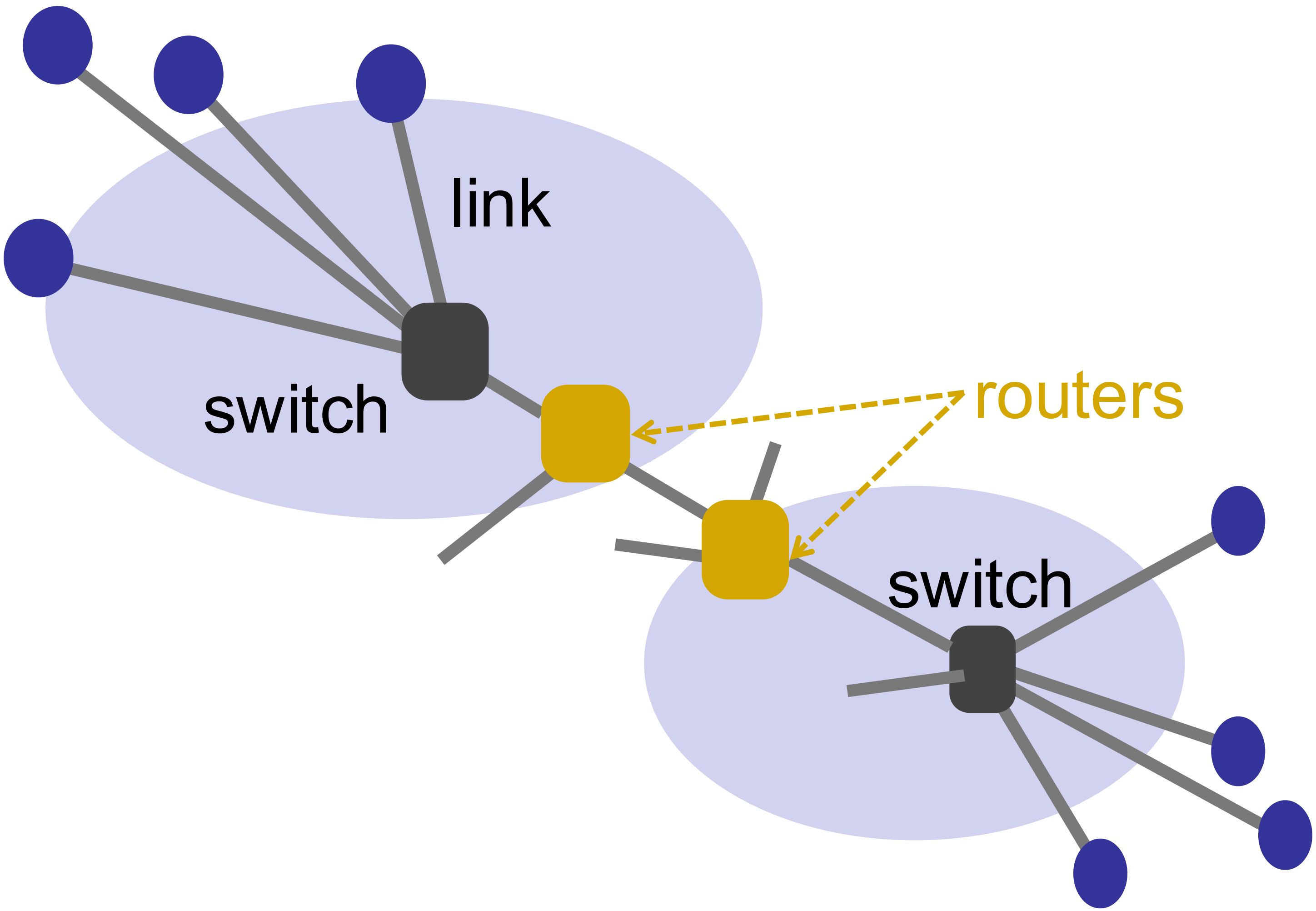
- Bits arrive on wire → physical layer (L1)
- Packets must be delivered across links and local networks → datalink layer (L2)
- Packets must be delivered between networks for global delivery → network layer (L3)

- Switches implement only physical and datalink layers (L1, L2)
- Routers implement the network layer too (L1, L2, L3)

A closer look at the network



A closer look at the network

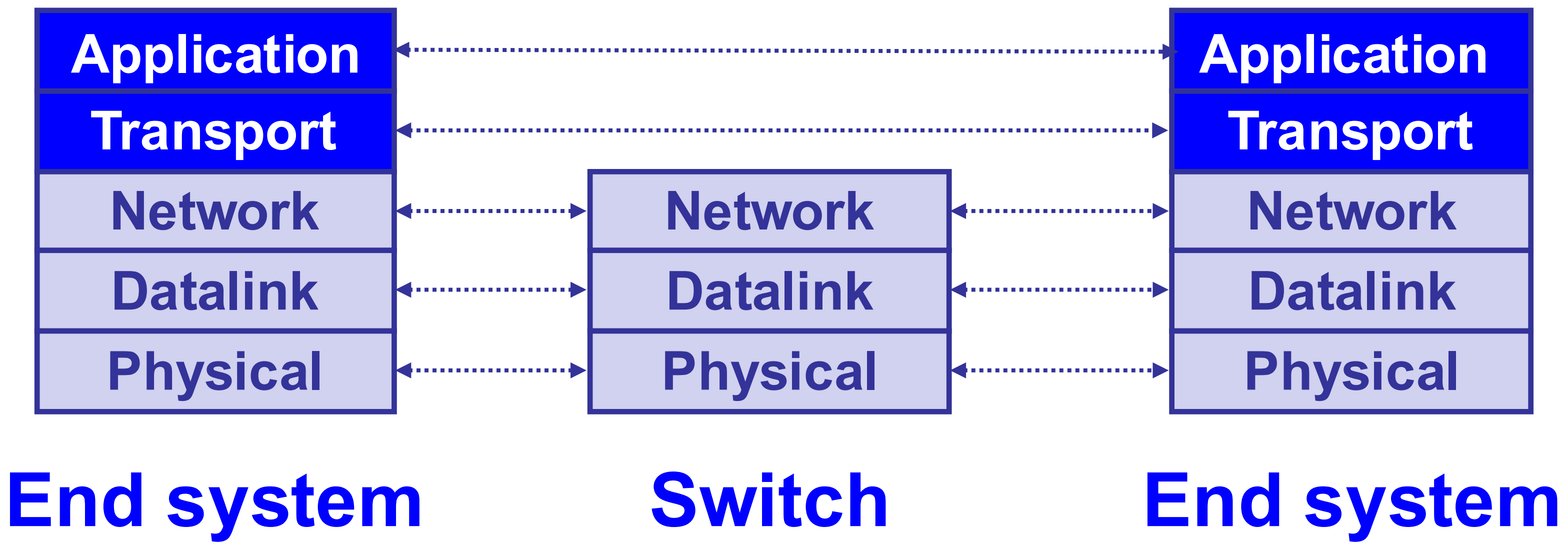


Switches vs. Routers

- Switches do what routers do but don't participate in global delivery, just local delivery
  - Switches only need to support L1, L2
  - Routers support L1-L3
- Won't focus on the router/switch distinction
  - Almost all boxes support network layer these days

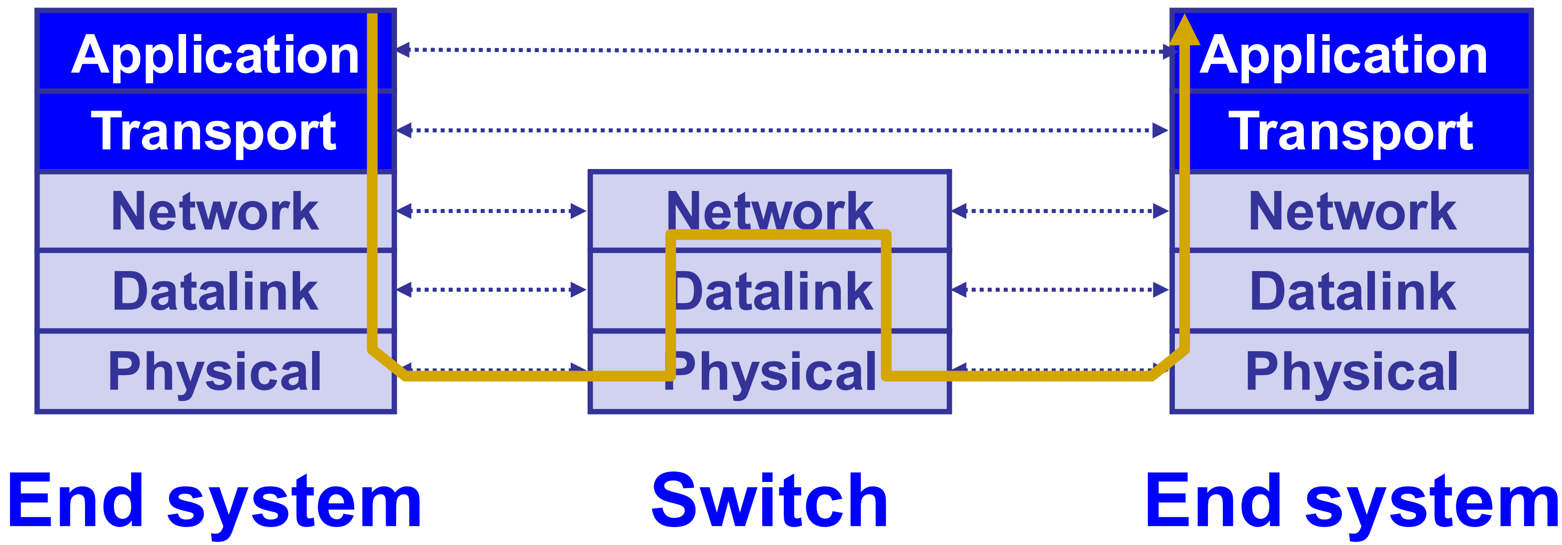
Logical communication

- A layer interacts with its peers in the corresponding layer

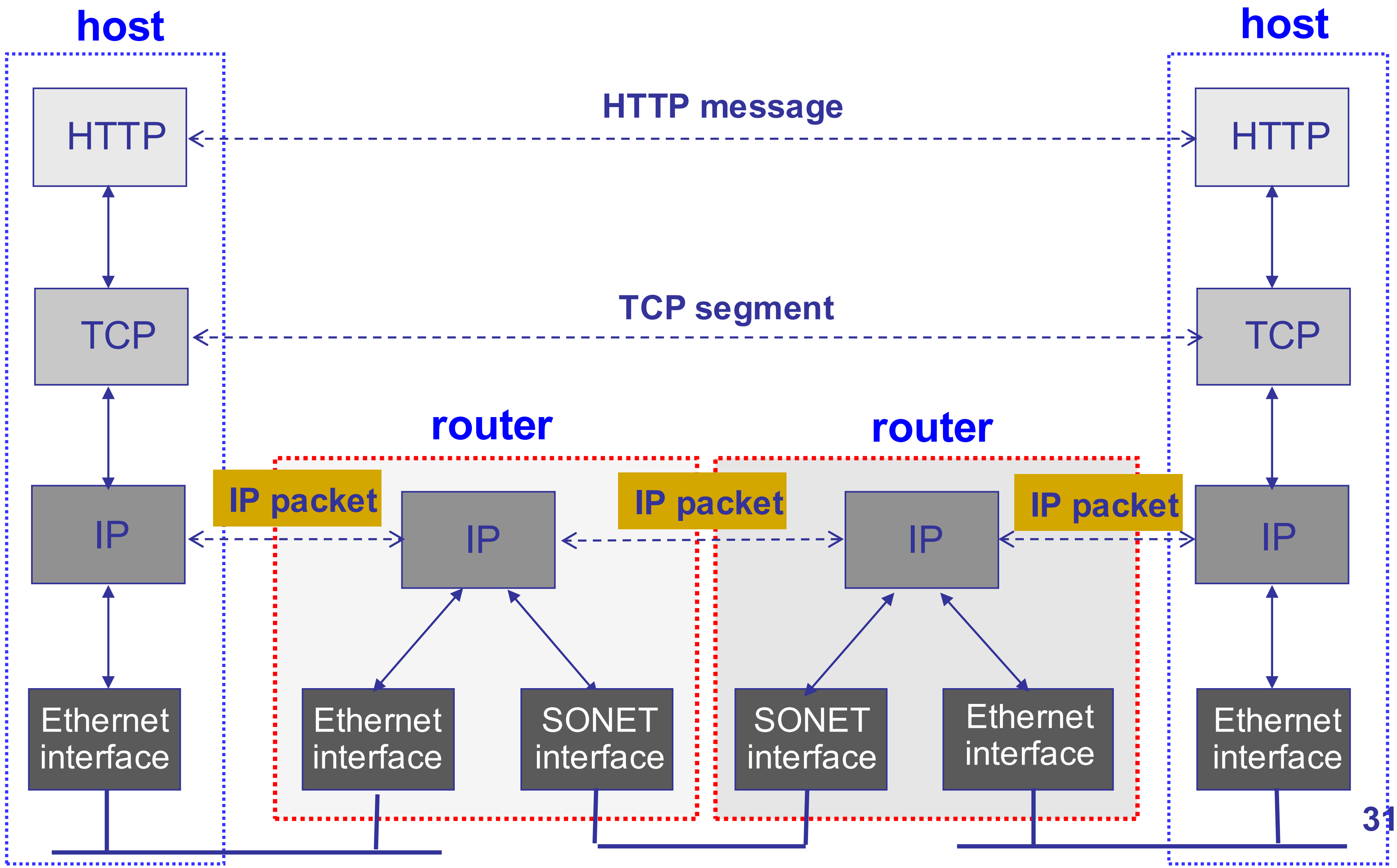


Physical communication

- Communication goes down to physical network
- Then up to relevant layer



A protocol-centric diagram



Pros and cons of layering

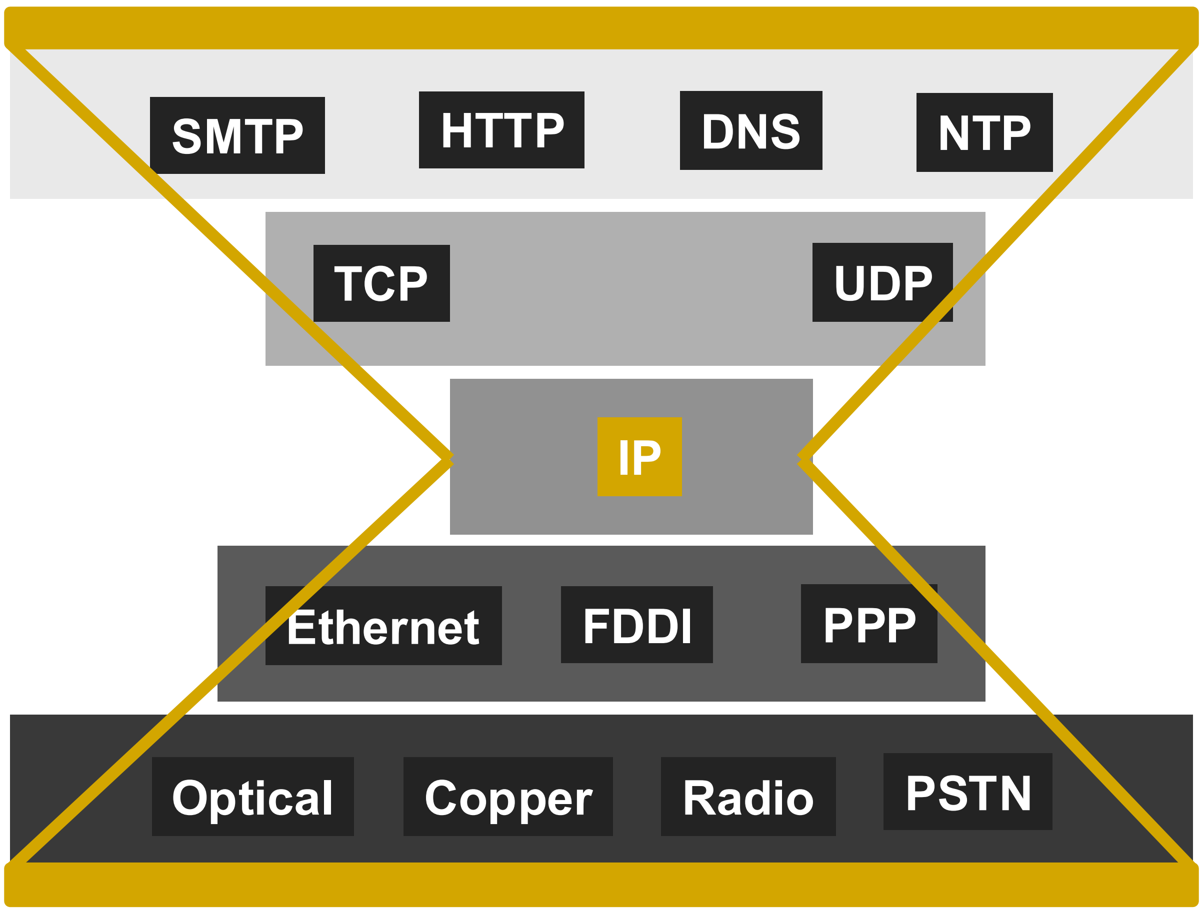
Why layers?

- Reduce complexity
- Improve flexibility

Why not?

- Higher overheads
- Cross-layer information often useful

IP is the narrow waist of the layering hourglass



Implications of hourglass

- Single network-layer protocol (IP)
- Allows arbitrary networks to interoperate
  - Any network that supports IP can exchange packets
- Decouples applications from low-level networking technologies
  - Applications function on all networks
- Supports simultaneous innovations above and below IP
- But changing IP itself is hard (e.g., IPv4 → IPv6)

Placing network functionality

- End-to-end arguments by Saltzer, Reed, and Clark
  - Dumb network and smart end systems
  - Functions that can be *completely* and *correctly* implemented *only* with the knowledge of application at end host, should not be pushed into the network
  - Sometimes necessary to break this for performance and policy optimizations
  - Fate sharing: fail together or don’t fail at all

Quick recap

- Layering is a good way to organize networks
- Unified Internet layer decouples applications from networks, allows technologies to evolve independently
- E2E argument encourages us to keep IP simple

TRANSPORT LAYER

Why a transport layer?

- IP addresses capture hosts, but end-to-end communication happens between applications
  - Need a way to decide which packets go to which applications (multiplexing/demultiplexing)
- IP provides a weak service model (best-effort)
  - Packets can be corrupted, delayed, dropped, reordered, duplicated
  - No guidance on how much traffic to send and when
  - Dealing with this is tedious for application developers

Multiplexing & demultiplexing

- Multiplexing (Mux)
  - Gathering and combining data chunks at the source from different applications and delivering to the network layer
- Demultiplexing (Demux)
  - Delivering correct data to corresponding sockets from a multiplexed stream

Role of the transport layer

- Communication between processes
  - Mux and demux from/to application processes
  - Implemented using *ports*

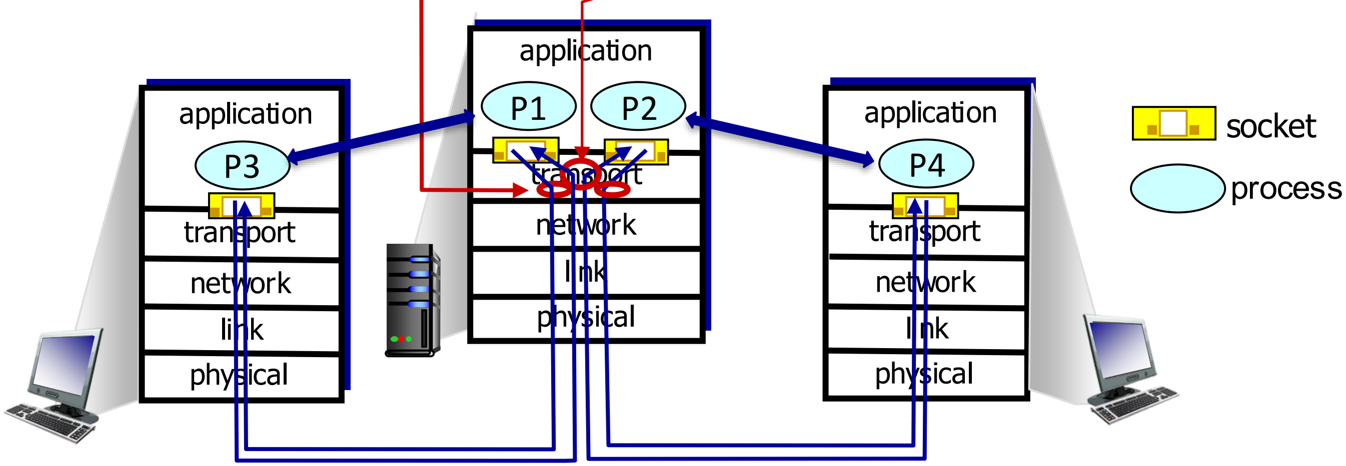
Multiplexing & demultiplexing

multiplexing at sender:

handle data from multiple sockets, add transport header (later used for demultiplexing)

demultiplexing at receiver:

use header info to deliver received segments to correct socket



Role of the transport layer

- Communication between processes
- Provide common end-to-end services for app layer [optional]
  - Reliable, in-order data delivery
  - Well-paced data delivery
    - Too fast may overwhelm the network
    - Too slow is not efficient

Role of the transport layer

- Communication between processes
- Provide common end-to-end services for app layer [optional]
- TCP and UDP are the common transport protocols
  - Also SCTP, MPTCP, SST, RDP, DCCP, …

Role of the transport layer

- Communication between processes
- Provide common end-to-end services for app layer [optional]
- TCP and UDP are the common transport protocols
- UDP is a minimalist transport protocol
  - Only provides mux/demux capabilities

Role of the transport layer

- Communication between processes
- Provide common end-to-end services for app layer [optional]
- TCP and UDP are the common transport protocols
- UDP is a minimalist transport protocol
- TCP offers a reliable, in-order, byte stream abstraction
  - With congestion control, but w/o performance guarantees (delay, b/w, etc.)

Applications and sockets

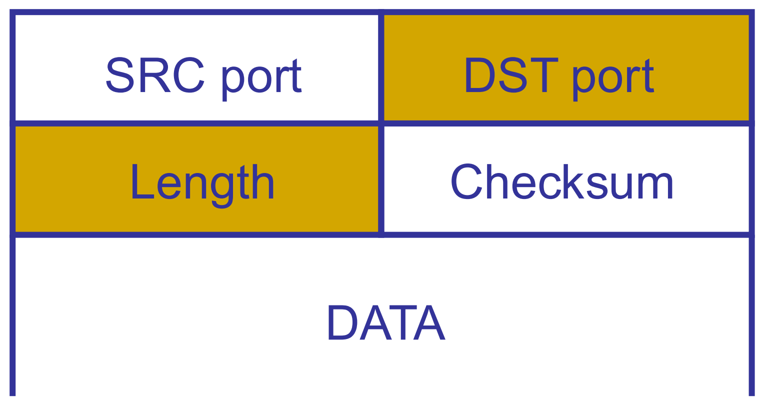
- Socket: software abstraction for an application process to exchange network messages with the (transport layer in the) operating system
- Two important types of sockets
  - UDP socket: TYPE is SOCK_DGRAM
  - TCP socket: TYPE is SOCK_STREAM

Ports

- 16-bit numbers that help distinguishing apps
  - Packets carry src/dst port no in transport header
  - Well-known (0-1023) and ephemeral ports
- OS stores mapping between sockets and ports
  - Port in packets and sockets in OS
  - For UDP ports (SOCK_DGRAM)
    - OS stores (local port, local IP address) ←→ socket
  - For TCP ports (SOCK_STREAM)
    - OS stores (local port, local IP, remote port, remote IP) ←→ socket

UDP: User Datagram Protocol

- Lightweight communication between processes
  - Avoid overhead and delays of order & reliability
- UDP described in RFC 768 – (1980!)
  - Destination IP address and port to support demultiplexing



UDP (cont'd)

- Optional error checking on the packet contents
  - (checksum field = 0 means "don't verify checksum")
- Source port is also optional
  - Useful to respond back to the sender in some cases

Why a transport layer?

- IP packets are addressed to a host but end-to-end communication is between application processes at hosts
  - Need a way to decide which packets go to which applications (mux/demux)
- IP provides a weak service model (best-effort)
  - Packets can be corrupted, delayed, dropped, reordered, duplicated
  - No guidance on how much traffic to send and when
  - Dealing with this is tedious for application developers

Reliable transport

- In a perfect world, reliable transport is easy

@Sender

- Send packets

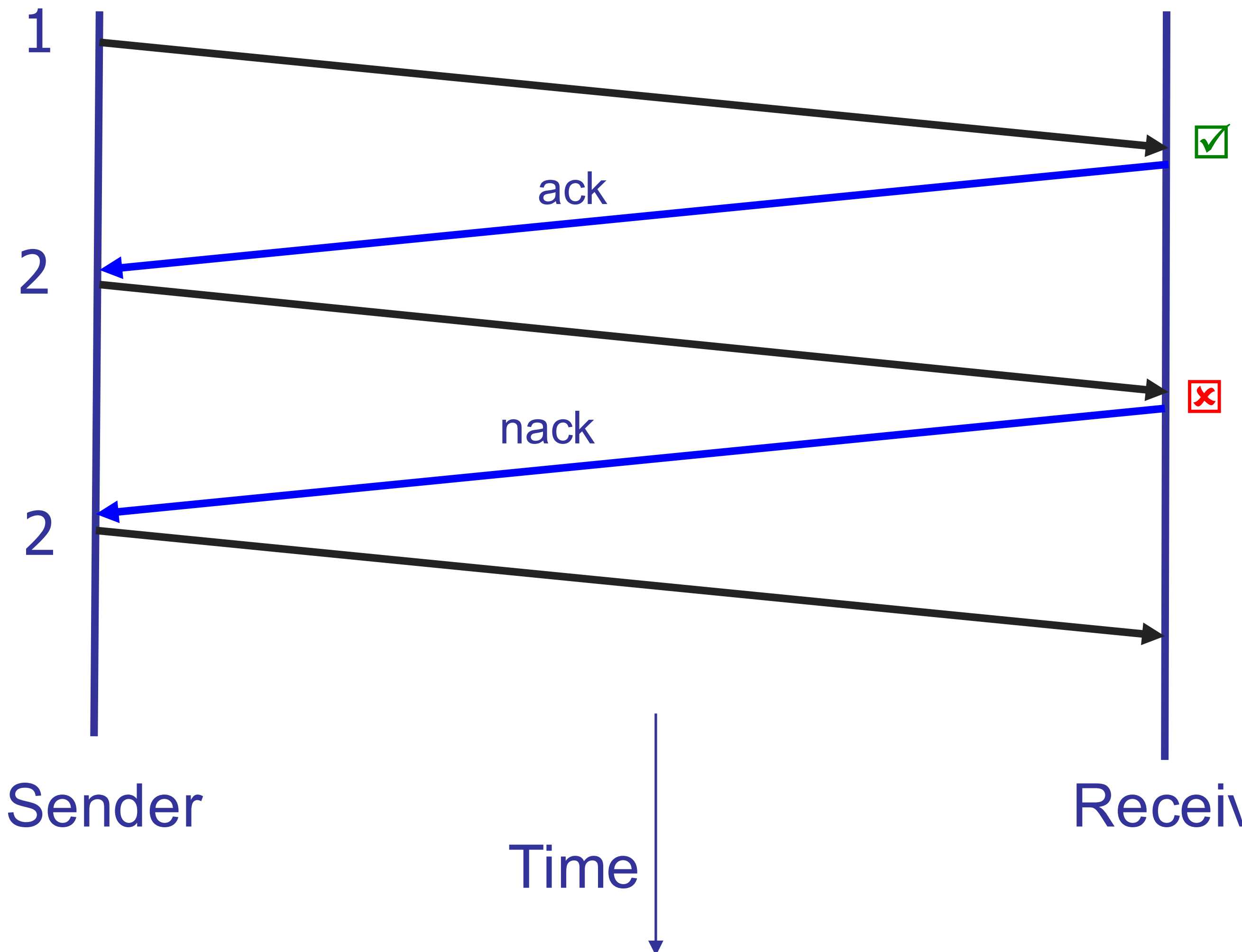
@Receiver

- Wait for packets

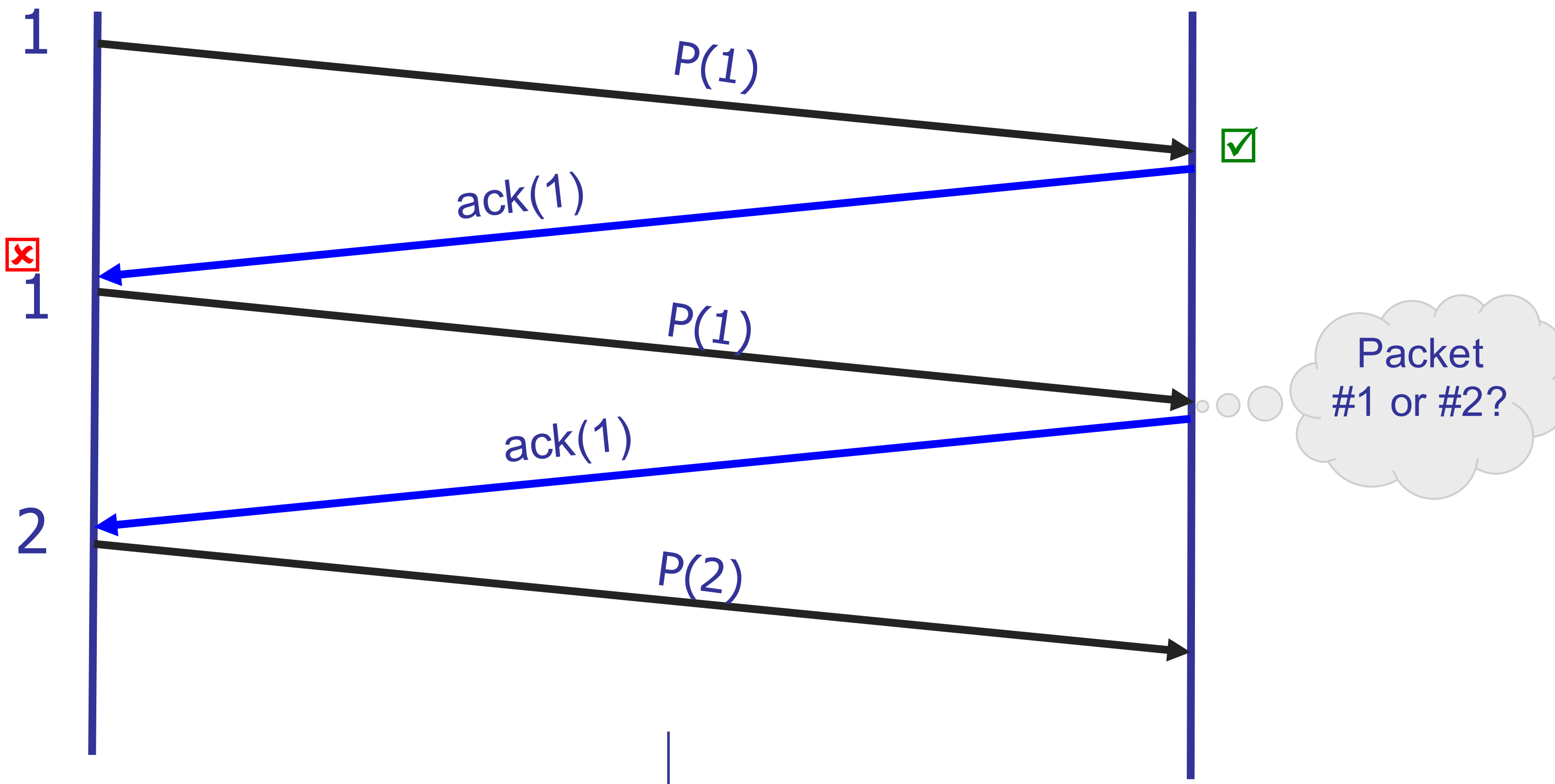
Reliable transport

- In a perfect world, reliable transport is easy
- All the bad things best-effort can do…
  - A packet is corrupted (bit errors)
  - A packet is lost (*why?*)
  - A packet is delayed (*why?*)
  - Packets are reordered (*why?*)
  - A packet is duplicated (*why?*)

Dealing with packet corruption

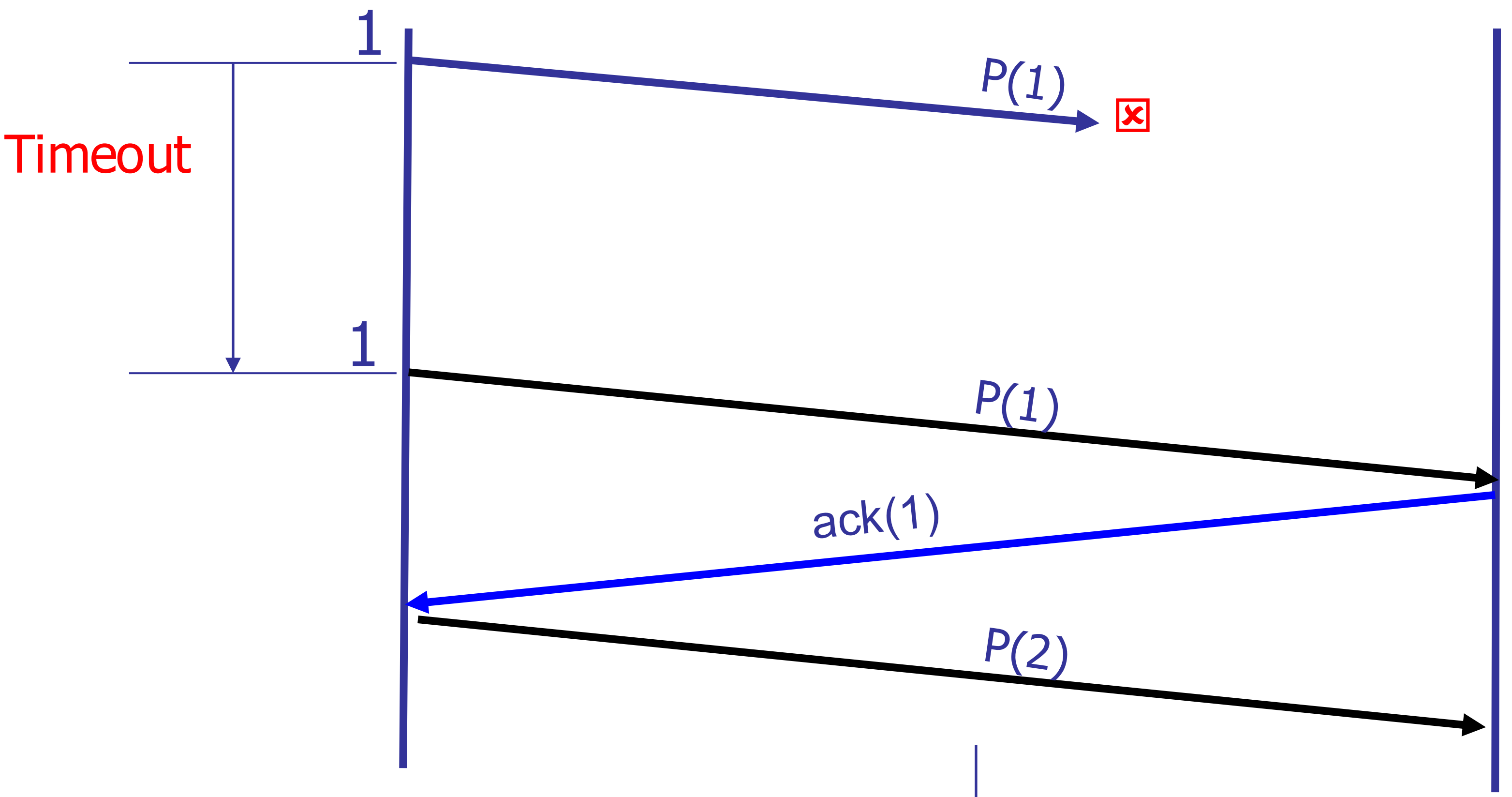


Dealing with packet corruption



Data and ACK packets carry sequence numbers

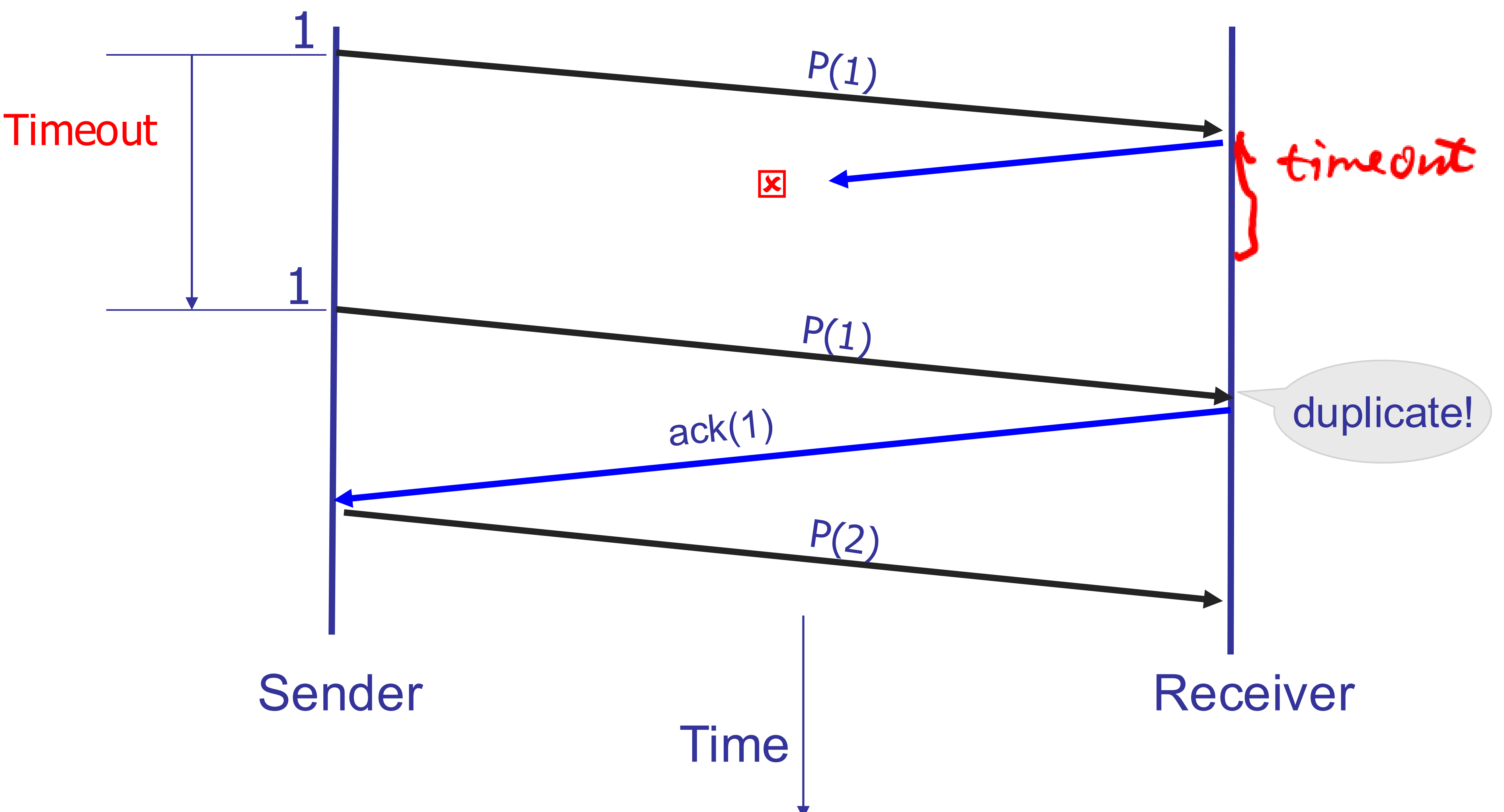
Dealing with packet loss



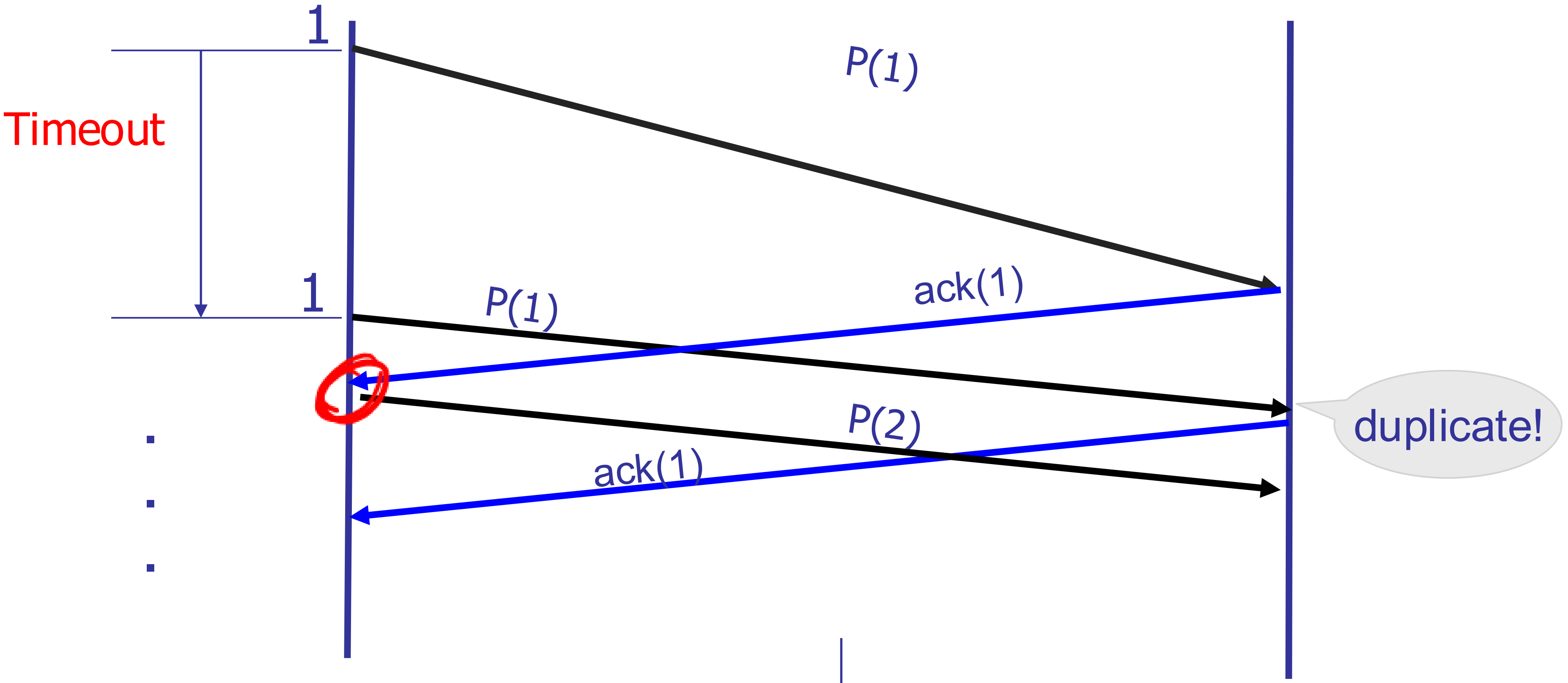
Timer-driven loss detection

Set timer when packet is sent; retransmit on timeout

Dealing with packet loss (of ack)



Dealing with delay



Timer-driven retransmission can lead to duplicates

Components of a solution

- Checksums (to detect bit errors)
- Timers (to detect loss)
- Acknowledgements (positive or negative)
- Sequence numbers (to deal with duplicates)

Quick recap

- Transport layer allows applications to communicate with each other
- Provides unreliable and reliable mechanisms
- Possible to build reliable transport over unreliable medium

Summary

- We’ve skipped many important materials
  - TCP: flow control, congestion control
  - L3 routing
  - User-space network stacks
  - L7 applications
  - Network management
- Networking is lots of fun!
  - Also lots of mess
- Take CSCI4430 next term if this excites you